

# Barrow $\Delta \leq 1$ from a long-range entanglement network: a capacity theorem and an area–volume dichotomy

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**Abstract.** We model the degrees of freedom of a gravitational horizon as a network of Planck-density sites, each with a bounded total entanglement weight  $\kappa$  (a finite local Hilbert-space dimension) and bond weights decaying with separation as  $w(r) \propto r^{-\alpha}$ . Three results follow. First, the bond-cut capacity of any region  $\Omega$  obeys  $C(\Omega) \leq \kappa|\Omega|$ , so the Barrow entropy deformation satisfies  $\Delta \leq 1$  as a theorem of the model (the *bound*; attaining  $\Delta = 1$  additionally needs the  $\alpha < d$  volume-basin geometry — derived here with an explicit finite-size error bound, uniform in position — and a cut-saturating horizon *state*, which we identify as the tracial state of the site algebra, leaving open only whether a physical horizon is driven to it) — no fractal-horizon geometry is needed, and for long-range couplings ( $\alpha < d$ ) none is available, since the minimum cut is not a surface. Second, the capacity is the enclosed *volume* ( $\Delta = 1$ ) for every range  $\alpha < d$  and the boundary *area* ( $\Delta = 0$ ) for  $\alpha > d + 1$ ; gravity’s Newtonian reach ( $\alpha = 1$ ) lies deep in the volume phase, and no gravitational coupling produces an intermediate range, so a measured intermediate  $\Delta$  falsifies the picture rather than selecting a parameter. Third,  $\Delta = 1$  exactly is the infrared-cutoff endpoint  $L \rightarrow \infty$  — a stated choice, not a derivation. What the network counts is a structural capacity (not manifestly the generalized entropy), and the volume identification is the non-default, empirically-bet-on branch: DESI DR3 + Euclid decide  $\Delta = 1$  versus 0 at a forecast  $\sigma(\Delta) \approx 0.09$ . This shortens — it does not eliminate — the assumption list behind the volume-law postulate of Structural Entropy Dark Energy. The results of §2 are self-contained and can be assessed on their own: the capacity theorem, the dichotomy and the crossover are statements about the network model of §2.1 alone, are state-independent, and take no input from the companion papers cited here, which supply the cosmological motivation for the question and the state half of its interpretation.

**Keywords:** holographic dark energy; horizon thermodynamics; entanglement entropy; de Sitter holography; Barrow entropy; long-range systems; apparent horizon; dark energy

## 1. Introduction

Structural Entropy Dark Energy (SEDE) reproduces the cosmological data with no fitted dark-sector parameter, at the cost of a single foundational input: the cosmic apparent horizon carries *volume-law* entropy,  $S \propto V$  (Barrow deformation  $\Delta = 1$ ), rather than the Bekenstein–Hawking area law. The cosmology paper [1] fixes the background from this input and confronts it with the data; the foundations paper [2] asks how much of the input is irreducible, decomposing it into *state* (the horizon degrees of freedom are maximally entangled), *form* (a thermalised state has volume-law entanglement), *scale* (the Cohen–Kaplan–Nelson bound fixes  $\rho_{\text{DE}} \sim \rho_{\text{crit}}$ ), and *count* (whether the entropy counts the enclosed *bulk* degrees of freedom or the *boundary* ones). The first three reduce to established physics; the count — bulk versus boundary — is the residue the foundations paper names as irreducible, a de Sitter holography question with no shortcut past the observer-algebra trace calibration.

This paper reduces the count to a shorter list of stated assumptions; it does not fully discharge it. We model the horizon degrees of freedom as a network with a finite per-site *leg budget*  $\kappa$  — equivalently, a finite local Hilbert-space dimension — and bond weights that decay as a power,  $w(r) \propto r^{-\alpha}$ . From this single, weaker premise three results follow (Figure 1), each machine-checked (§Reproducibility):

1.  $\Delta \leq 1$  is a theorem (§2.2). The bond-cut capacity of any region obeys  $C(\Omega) \leq \kappa \cdot |\Omega|$  identically — the entropy a region can carry cannot exceed its degree-of-freedom count. No fractal ceiling and no premise that the horizon is a space-filling surface are needed;  $\Delta > 1$  is impossible, and a high-significance  $\Delta > 1$  would falsify the framework with no parameter to absorb it.
2. The count is binary, and gravity is on the volume side (§2.3–§2.4). The capacity exponent is the enclosed *volume* ( $\Delta = 1$ ) for every coupling of range  $\alpha < d$  and the *area* ( $\Delta = 0$ ) for  $\alpha > d + 1$ , with the crossover at  $\alpha = d$  exactly — the same threshold that separates additive from non-additive long-range statistical mechanics and at which the entanglement area-law theorems [3] lose their hypothesis (§3). The  $\alpha$ -controlled volume→(fractal)→area crossover itself is established in the long-range-entanglement literature [4–6] and, for the network cut-scaling itself, in long-range percolation and small-world graph theory [7–9]; our contribution is to read a Barrow index  $\Delta = q - 2$  off it for a *horizon* count, not the crossover. Gravity’s  $1/r$  reach sits at  $\alpha = 1$ , deep in the volume basin (§4); the equilibrium, area-law frameworks are its local limit. Whatever selects  $\Delta = 1$  must therefore be a *saturation* argument.
3.  $\Delta = 1$  as the infinite-cutoff endpoint (§5). The only source of a deficit below the volume law is a finite infrared cutoff  $L$  on the network; nothing *internal* to SEDE fixes  $L$ , so  $\Delta$  is a monotone function of it, and  $\Delta = 1$  is the  $L \rightarrow \infty$  endpoint suggested by the flat, infinite spatial slice. This is a *choice of cutoff* — the identification of the cosmological spatial slice with the entanglement-network infrared cutoff — not a derivation (§5.3–§5.4);  $\Delta = 1$  remains the theory’s one postulate, restated at the endpoint.

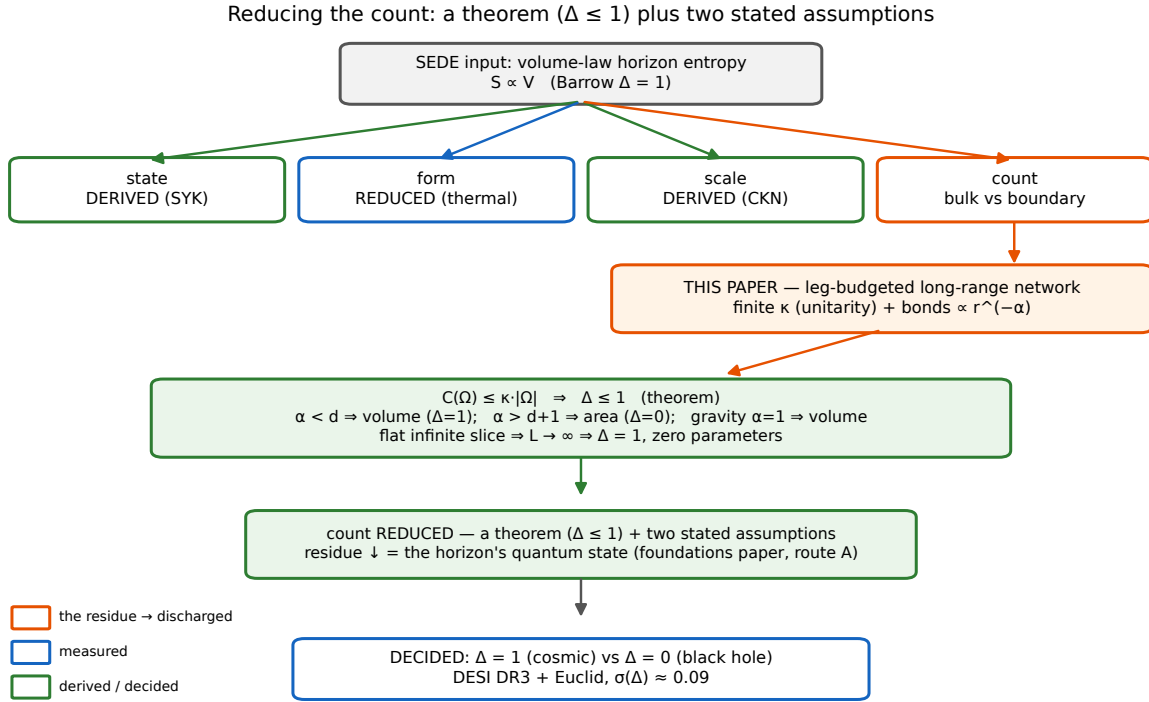


Figure 1. The reduction. SEDE’s one input — volume-law horizon entropy — decomposes into *state*, *form*, *scale*, and *count* (the foundations paper); the first three reduce to established physics, and the *count* is the irreducible residue. Modelling the horizon degrees of freedom as a leg-budgeted long-range network reduces it:  $C(\Omega) \leq \kappa \cdot |\Omega|$  makes  $\Delta \leq 1$  a theorem (§2.2),  $\alpha < d$  puts gravity on the volume branch (§2.3–§4), and  $\Delta = 1$  is the  $L \rightarrow \infty$  endpoint of the flat infinite slice — a cutoff choice, not a derivation (§5). What survives is the horizon’s quantum *state* (route A of the foundations paper) and that endpoint identification, decided — with the black-hole  $\Delta = 0$  — by DESI DR3 + Euclid.

The residue that survives is *strictly weaker* than the postulate it replaces (§7): a finite leg budget (unitarity), a bond weight decaying slower than  $r^{-d}$  (an inequality, not a functional form), the infinite-cutoff identification  $L \rightarrow \infty$  (a stated choice, §5.3), and the maximally-scrambled state of route A of the foundations paper (untouched). What is bought is that the open content of SEDE now sits in two named places — the horizon’s quantum *state* and the  $L \rightarrow \infty$  endpoint reading of its *count* — and, for the one horizon we have, is decided by a scheduled measurement:  $\Delta = 1$  for the driven cosmic horizon and  $\Delta = 0$  for equilibrium (black-hole) horizons, separated by DESI DR3 + Euclid at a forecast  $\sigma(\Delta) \approx 0.09$ . We are explicit about what we do not claim: the network is a *model* of the horizon degrees of freedom, not a derivation of them from a fundamental theory, and the bond cut is a *capacity*, an upper bound whose attainment is the state question (§6). What the paper delivers is the counting residue reduced to a theorem (the ceiling) plus two honestly-stated assumptions (the network model with its range inequality, and the  $L \rightarrow \infty$  endpoint choice), and a geometry — the count is not a Hausdorff dimension — sharp enough to be falsified.

## 2. The count from a leg-budgeted long-range network

### 2.1. Setup

Let the horizon degrees of freedom be sites of a network filling a ball of radius  $L$  at number density  $n = \ell_P^{-d}$ ,  $d = 3$ , with symmetric bond weights

$$w_{ij} \propto r_{ij}^{-\alpha}, \quad \sum_j w_{ij} = \kappa \quad \text{for every } i. \quad (2.1)$$

The budget in (2.1) is not an idealisation. In a tensor-network representation the leg weight attached to a site is  $\ln(\text{local Hilbert-space dimension})$ ; unitarity bounds it. Given the shape  $r^{-\alpha}$ , the budget is imposed by the unique symmetric scaling  $w = \text{diag}(d) W \text{diag}(d)$  with prescribed row sums — the symmetric Sinkhorn problem, which for positive symmetric  $W$  has a unique positive solution. All numbers below are computed at Sinkhorn residual  $< 10^{-9}$ .

We take as the count the bond-cut capacity of a region  $\Omega$ ,

$$C(\Omega) = \sum_{i \in \Omega, j \notin \Omega} w_{ij}, \quad (2.2)$$

the standard upper bound on the entanglement entropy of  $\Omega$  with its complement.  $C$  is a capacity, not an entropy: whether it is attained depends on the state (§6). Everything in this section concerns  $C$ .

### 2.2. The capacity theorem

From (2.1) and (2.2), for any region  $\Omega$ ,

$$C(\Omega) \leq \sum_{i \in \Omega} \sum_j w_{ij} = \kappa |\Omega|. \quad (2.3)$$

The entropy of a region cannot exceed its degree-of-freedom count. Writing  $S \propto R^{2+\Delta}$  in the Barrow parametrisation  $S = (A/A_0)^{1+\Delta/2}$ , (2.3) gives

$$\Delta \leq 1. \quad (2.4)$$

This is the ceiling of the cosmology paper (§8.1) [1] — obtained here without reference to the Hausdorff dimension of the horizon, and without the premise that the horizon is a fractal 2-surface. Finite local dimension is the whole content. The theorem is model-conditional: it is a statement about the leg-budgeted network of §2.1 — with the  $\alpha = d$  crossover of §2.3 now carrying an explicit finite-size error bound (Proposition 2.1) rather than an unproven limit interchange — not about an arbitrary horizon. Two consequences are worth stating plainly. First,  $\Delta > 1$  is impossible in this framework, so a

high-significance measurement of  $\Delta > 1$  falsifies it with no parameter available to absorb the result. Second, whatever selects  $\Delta = 1$  must be a *saturation* argument, since (2.3) is the only bound in play.

### 2.3. The crossover is at $\alpha = d$

Consider a site at radius  $s$  inside a sphere of radius  $\rho \ll L$ . The bond mass it places at separations in  $[u, u+du]$  scales as  $u^{d-1-\alpha} du$ . Its outward fraction — the share of its budget spent on bonds crossing  $\rho$  — is therefore governed by

$$\int^L u^{d-1-\alpha} du \quad \text{versus} \quad \int^\rho u^{d-1-\alpha} du. \quad (2.5)$$

For  $\alpha < d$  the integral is dominated by its upper limit: the bond mass sits at the largest separations, the outward fraction tends to unity for every interior site, and

$$C(\rho) \rightarrow \kappa n V(\rho) \quad (\text{volume law, budget saturated}). \quad (2.6)$$

As stated, (2.6) is a limiting statement, and the limits do not commute: at fixed ambient size  $L$  the outward fraction *decreases* with  $\rho$  and must vanish as  $\rho \rightarrow L$ , so “dominated by the upper limit” cannot be read as  $\rho \rightarrow \infty$  at fixed  $L$ . The following proposition replaces the limit by an explicit finite-size error, uniform in the site position, and is what the volume law rests on here.

**Proposition 2.1 (uniform outward-fraction bound).** Let sites of density  $n = \ell^{-d}$  fill a ball of radius  $L$ , let  $\alpha < d$ , and let  $f_{\text{out}}(x)$  be the fraction of the budget of a site at  $x$ ,  $|x| \leq \rho$ , carried by bonds crossing the sphere of radius  $\rho$ . Then for every such  $x$ ,

$$f_{\text{out}}(x) \geq 1 - K(d, \alpha) (\rho/L)^{d-\alpha}, \quad K(d, \alpha) = \frac{S_{d-1} 2^{d-\alpha} / (d - \alpha)}{c_d (1 - 2^{-d}) 2^{-\alpha}},$$

with  $S_{d-1}$  the unit-sphere surface and  $c_d$  the unit-ball volume. Consequently

$$\kappa n V(\rho) [1 - K(d, \alpha) (\rho/L)^{d-\alpha}] \leq C(\rho) \leq \kappa n V(\rho). \quad (2.6a)$$

*Proof.* Every point of the interior ball lies within distance  $2\rho$  of  $x$ , so the inward mass is at most  $n \int_0^{2\rho} S_{d-1} u^{d-1-\alpha} du = n S_{d-1} (2\rho)^{d-\alpha} / (d - \alpha)$ , the integral converging at  $u = 0$  precisely because  $\alpha < d$ . Every point of the outer shell  $L/2 < |y| < L$  lies within distance  $2L$  of  $x$ , so the outward mass is at least  $n c_d (1 - 2^{-d}) L^d \cdot (2L)^{-\alpha}$ . Since  $f_{\text{out}} = 1 / (1 + \text{in/out}) \geq 1 - \text{in/out}$ , the ratio of these two gives the stated  $K$ . The upper bound in (2.6a) is the capacity theorem of §2.2. ■

Three things follow. First, the volume law is a statement at finite  $(\ell, \rho, L)$  with a controlled error, so no interchange of  $\rho \rightarrow \infty$  with the continuum limit is invoked. Second, the same condition  $\alpha < d$  that makes the short-distance integral converge is what

makes the discrete-to-continuum replacement safe: the near-field discrepancy between the lattice sum and the integral is relatively  $O((\ell/\rho)^{d-\alpha})$ , vanishing in the same regime. Third, the error term makes explicit *why*  $\Delta = 1$  exactly is the  $L \rightarrow \infty$  endpoint of §5 rather than a finite-size result — the deficit is governed by  $\rho/L$ , not by  $\rho$  alone.

The bound is rigorous but its constant is not sharp. Numerically (src/outward\_fraction\_bound.py,  $d = 3$ ,  $L = 200$ , exact angular average), the measured capacity deficit follows the predicted power  $(\rho/L)^{d-\alpha}$  with a constant of order unity: at gravity's  $\alpha = 1$  the fitted deficit exponent is 2.03 against the predicted  $d - \alpha = 2$ , with  $K_{\text{measured}} \approx 0.79$  stable across  $\rho \in [5, 40]$ , while the proved constant is  $K = 13.7$  — valid, and loose by a factor  $\approx 17$ . The fit degrades as  $\alpha \rightarrow d$  (at  $\alpha = 2.5$  the fitted exponent is 0.79 against a predicted 0.5), which is expected: there the deficit is no longer small and the leading term does not yet control the behaviour. For gravity's range the asymptotics are firmly in their regime.

For  $\alpha > d$  the integral converges at small  $u$ : only a surface layer of thickness  $O(\ell_p)$  contributes, and  $C(\rho) \propto A(\rho)$  (area law). The crossover is at  $\alpha = d$  exactly — the same threshold that separates additive from non-additive long-range systems (Campa–Dauxois–Ruffo [10]), and the same one at which the hypothesis of the entanglement area-law theorems (Hastings [11]; Brandão–Horodecki [12]) fails.

Numerical confirmation (Figure 2). Spherical reduction with the exact angular average of  $|x - y|^{-\alpha}$ , ambient  $L = 200$ ,  $\kappa = 1$ . The table gives  $s(\rho) \equiv C(\rho)/N_{\text{in}}$ , the capacity per enclosed degree of freedom, and  $q$  from  $C \propto \rho^q$ :

Table 1: Capacity per enclosed degree of freedom  $s(\rho) = C(\rho)/N_{\text{in}}$  and measured capacity exponent  $q$  ( $C \propto \rho^q$ ) versus bond range  $\alpha$ , in the spherical reduction ( $L = 200$ ,  $\kappa = 1$ ).

| $\alpha$ | $\rho = 2$ | $\rho = 5$ | $\rho = 10$ | $\rho = 20$ | $\rho = 40$ | $q$  |
|----------|------------|------------|-------------|-------------|-------------|------|
| 1.0      | 0.9999     | 0.9996     | 0.9983      | 0.9933      | 0.9728      | 2.99 |
| 2.0      | 0.9969     | 0.9873     | 0.9707      | 0.9371      | 0.8690      | 2.96 |
| 2.5      | 0.9805     | 0.9375     | 0.8850      | 0.8089      | 0.6994      | 2.89 |
| 4.0      | 0.7088     | 0.4156     | 0.2569      | 0.1518      | 0.0863      | 2.30 |
| 6.0      | 0.5135     | 0.2206     | 0.1119      | 0.0562      | 0.0281      | 2.03 |

At  $\alpha = 6$  the capacity per site halves exactly when  $\rho$  doubles:  $s \propto 1/\rho$ , the area law, to three figures. At  $\alpha = 1$  it sits at  $\kappa$  to four decimals: every degree of freedom spends essentially its entire budget outward. The volume law is not imposed by a ceiling; it is attained.

The  $\alpha = 4$  row warrants a word, because its  $q = 2.30$  has not reached 2 and could be mistaken for a failure to converge. It is not:  $\alpha = 4 = d + 1$  is the *marginal* area point, where the capacity-carrying surface layer has a logarithmic tail,  $C(\rho) \propto \rho^2 \cdot \ln \rho$ , so the finite- $\rho$  exponent is  $q(\rho) = 2 + O(1/\ln \rho)$  — inflated above 2 but *decreasing*, not a stable

power. Pushing the ambient size out ( $L = 1500$ ,  $\rho$  up to 180) settles it with a normalisation-free diagnostic,  $C(\rho)/\rho^2$ : at  $\alpha = 4$  it rises linearly in  $\ln \rho$  (*slope*  $\approx 2.9 > 0$ ), the signature of the marginal logarithm, whereas at  $\alpha = 5$  — just past the marginal point — it is flat to 6 % with  $q = 2.01$ , the clean area law. No  $\alpha$  produces a *stable* exponent between 2 and 3: the  $\alpha = 4$  elevation is a marginal logarithm on the area side, and it sharpens rather than softens the binary dichotomy of §2.4. (For  $d < \alpha < d + 1$  the same analysis gives a genuine but still non-volume  $q \rightarrow 3 - (\alpha - d)$ ; at  $\alpha = 3.5$ ,  $q$  climbs toward 2.5. The volume phase,  $q = 3$ , is reached only for  $\alpha < d$ .)

Identifying the saturated capacity with a constant entropy density,

$$s_0 = \kappa n = \kappa/\ell_P^3, \quad (2.7)$$

recovers eq. (2.2) of the cosmology paper [1],  $s_{\text{grav}} = s_0$ , with the constant identified as legs per Planck cell.

#### 2.4. The dichotomy is binary — for a gravitational horizon

Equation (2.6) holds for every  $\alpha < d$ , not merely  $\alpha = 1$ : the volume phase is a basin, not a knife-edge. The model therefore never requires the specific form  $w(r) = 1/r$ . It requires only

$$w(r) \text{ decays slower than } r^{-d}. \quad (2.8)$$

Any coupling of gravitational reach satisfies (2.8) with room to spare. We are careful about what is and is not binary. The capacity exponent is  $q = 3$  (volume) for *every*  $\alpha < d$  and  $q = 2$  (area) for  $\alpha > d + 1$ ; the intermediate window  $d < \alpha < d + 1$  does interpolate,  $q = 3 - (\alpha - d)$ , with  $\alpha = d + 1$  the marginal area case ( $q \rightarrow 2$  with a logarithm) — the crossover is a genuine continuum in  $\alpha$  (§2.3), not a two-valued function. What is binary is the count for a *gravitational* horizon: gravity is  $1/r$ , so  $\alpha = 1 \ll d$ , sitting deep in the volume basin, while the equilibrium/area frameworks are local (effectively  $\alpha > d + 1$ ). An intermediate  $\Delta$  would require an intermediate-*range* horizon coupling,  $d < \alpha < d + 1$ , for which there is no physical mechanism — gravity supplies no such exponent. A measured intermediate  $\Delta$  therefore falsifies the gravitational-network picture — no coupling of gravitational reach produces it — rather than selecting a tunable exponent. In particular it closes the door on the Kardar–Parisi–Zhang branch ( $\Delta \approx 0.61$ ) once floated as a distinguishable alternative in the foundations paper: §2.5 shows there is no roughening universality class to select in the first place.

#### 2.5. For $\alpha < d$ there is no minimum-cut surface

The Ryu–Takayanagi prescription reads entanglement entropy off a *minimal cut surface*. That object exists only when the bonds are local. We test this directly by maximum flow on a lattice ball, with boundary legs of the upper cap tied to a source and the lower cap



The binary count: gravity ( $\alpha=1$ ) saturates the volume; no gravitational coupling gives intermediate  $\Delta$

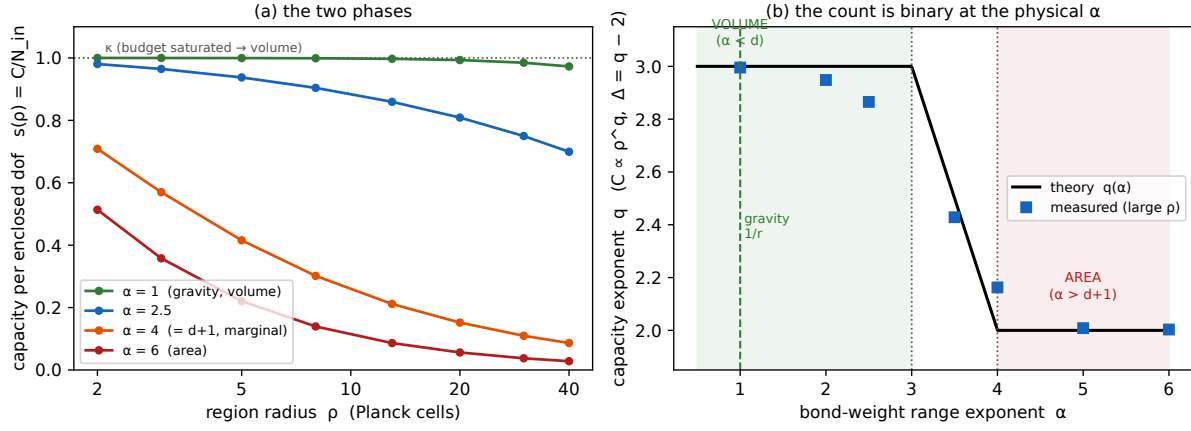


Figure 2. The binary count (§2.3–2.4). (a) Capacity per enclosed degree of freedom  $s(\rho) = C/N_{\text{in}}$ : at  $\alpha = 1$  (gravity) it saturates the leg budget  $\kappa$  (the volume law); at  $\alpha = 6$  it falls as  $1/\rho$  (the area law). (b) The capacity exponent  $q$  (with  $\Delta = q - 2$ ) versus the bond-range  $\alpha$ :  $q = 3$  (volume) for every  $\alpha < d$  and  $q = 2$  (area) for  $\alpha > d + 1$ , interpolating only in the *unphysical* window  $d < \alpha < d + 1$  (the marginal  $\alpha = d + 1 = 4$  point carries a logarithm, so its finite- $\rho$  value sits just above 2). Gravity’s  $1/r$  reach ( $\alpha = 1$ ) lies deep in the volume basin; no gravitational coupling yields an intermediate  $\Delta$ .

to a sink at infinite capacity, so that every cut passes through the bulk. Two diagnostics, neither of which depends on any normalisation:

(i) Thickness of the severed-bond set, in lattice units (weighted RMS of the midpoint height of cut bonds):

Table 2: Thickness of the severed-bond set (weighted RMS midpoint height, lattice units) versus ball radius  $R$  for each bond range  $\alpha$ , from the maximum-flow minimum cut on the lattice ball.

| $\alpha$ | R=3   | R=4   | R=5   | R=6   |
|----------|-------|-------|-------|-------|
| 0.5      | 0.78  | 1.20  | 1.60  | 1.98  |
| 1.0      | 0.82  | 1.27  | 1.70  | 2.11  |
| 2.0      | 0.86  | 1.43  | 1.92  | 2.39  |
| 3.0      | 0.57  | 1.54  | 2.11  | 2.68  |
| 4.0      | 0.55  | 0.59  | 0.63  | 0.64  |
| 6.0      | 0.516 | 0.516 | 0.520 | 0.518 |

For  $\alpha > d$  the thickness is  $R$ -independent — a sharp interface one lattice spacing thick, and the cut is the equatorial disk ( $|S|/N \rightarrow 0.447$ , a half-ball). For  $\alpha \leq d$  it grows  $\propto R$ : the severed bonds fill the volume (Figure 3).



No minimum-cut surface for a long-range horizon  $\Rightarrow$  no height field, no roughening (§2.5)

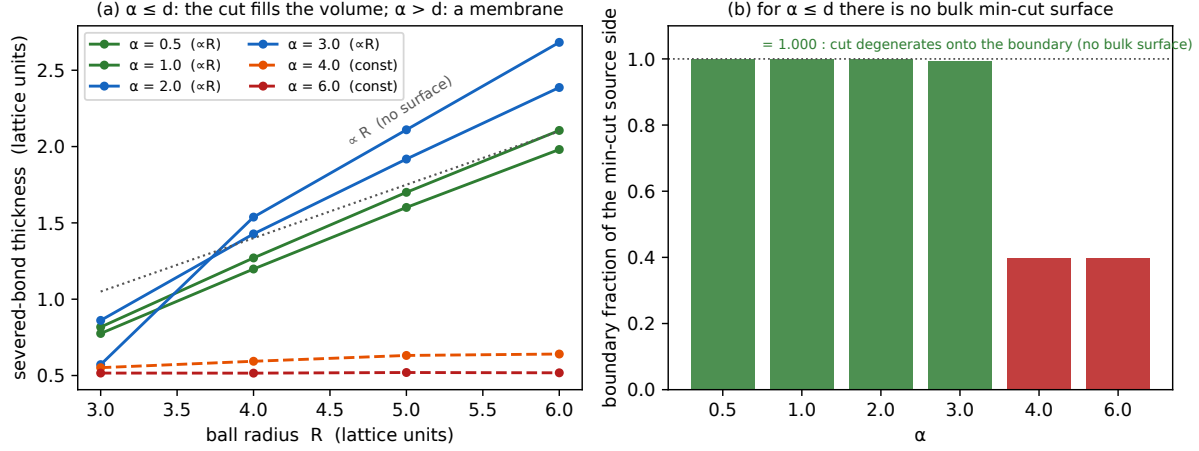


Figure 3. No minimum-cut surface for a long-range horizon (§2.5). (a) The thickness of the severed-bond set grows  $\propto R$  for  $\alpha \leq d$  (the cut fills the volume — there is no surface) and is  $R$ -independent for  $\alpha > d$  (a one-cell membrane). (b) The boundary fraction of the minimum cut's source side is 1.000 for  $\alpha \leq d$  (the cut degenerates onto the boundary shell) and  $\approx 0.40$  for  $\alpha > d$  (a genuine bulk membrane). With no min-cut surface there is no height field, no roughness exponent, and no roughening — so  $\Delta$  is a count, not a Hausdorff dimension.

(ii) Where the cut lives. For every  $\alpha \leq 3$  and every  $R$  tested, the fraction of the source side of the minimum cut consisting of boundary sites is 1.000, exactly: the cut degenerates onto the boundary shell. There is no bulk surface, because there is no emergent bulk geometry. (The cut cost then scales as  $R^{5-\alpha}$  — super-volume — which is the signature of the *unbudgeted* network's illegitimacy: per-site leg weight  $\sum_j r_{ij}^{-1} \sim R^2$  diverges, and no finite local Hilbert space supports it. Imposing (2.1) removes the pathology and returns (2.6).)

The consequence is structural. With no minimum-cut surface there is no height field, no roughness exponent, and no roughening universality class.  $\Delta$  is a count, not a Hausdorff dimension, exactly as the cosmology paper (§8.5) maintains. This is also the foundations paper's position: it reads the count as the bare bound  $\Delta \leq 1$ , which §2.2 here *derives* — so the fractal reading  $\Delta = d_H - 2$  is unnecessary.‡

An independent confirmation from the classical side: in linearised GR the apparent-horizon condition  $\theta_+ = 0$  is, in Newtonian gauge (where the constant-time slicing is shear-free, so  $K_{ij}$  is pure trace), the constant-mean-curvature condition  $\mathcal{M} = 2H$ . It is the Euler-Lagrange equation of  $A[h] - 2H \cdot V[h]$ , and since the area functional  $A = \bar{R}^2 \int d\Omega [(1+h)^2 + \frac{1}{2} |\nabla_S h|^2 + O(|\nabla_S h|^4)]$  carries no cubic term, the constraint reads

‡ The  $\epsilon$ -lag / roughening observable of the companion papers is superseded here: eq. (2.9) leaves the classical apparent horizon with no growth equation to roughen with. The companion papers already reflect this — the foundations paper reduces its membrane section to the exact coefficients and withdraws the roughening layer, and the cosmology paper withdraws the corresponding predictions (its §6).

$$\nabla_S^2 h + 2h + 2h^2 = 2\mathcal{S}, \quad \mathcal{S} = \Phi + \Psi + \dot{\Psi}/H - H^{-1}\partial_R\Psi. \quad (2.9)$$

There is no  $(\nabla h)^2$  term at second order, and — decisively — no  $\partial_t h$  at any order. The operator  $\nabla_S^2 + 2$  is elliptic on the sphere;  $h_{\ell m} = 2\mathcal{S}_{\ell m}/(2 - \ell(\ell + 1))$  is slaved instantaneously to the matter perturbations, and high multipoles are suppressed as  $1/\ell^2$ . The classical apparent horizon does not roughen. It has no growth equation to roughen with. That the apparent-horizon shape can be found this way — perturbatively, order by order, as a slaved constraint within the canonical formalism — is independently established by Neuser and Thiemann, who compute it to second order in gauge-invariant perturbation theory [13].

This operator is not ours to invent:  $\nabla_S^2 + 2$  is the Andersson–Mars–Simon marginally-outer-trapped-surface (MOTS) stability operator [14] specialised to the FRW apparent horizon. On the round 2-sphere (areal radius  $\bar{R} = 1/H$ ) the AMS operator  $L\psi = -\nabla_S^2\psi + 2s^A D_A\psi + (1/2 R_S - |\sigma^+|^2 - s_A s^A + D_A s^A - 8\pi T_{ab}\ell^a k^b)\psi$  collapses — spherical symmetry gives  $s_A = 0$ , the horizon is shear-free ( $\sigma^+ = 0$ ), and for the within-slice shape variation it reduces to the classical mean-curvature Jacobi operator  $-(\nabla_S^2 + |A|^2)$  with  $|A|^2 = 2/\bar{R}^2$ , i.e. exactly  $-(1/R)(\nabla_S^2 + 2)$ . Its spectrum is the published one,  $\ell(\ell + 1) - 2$ , and the  $\ell = 1$  kernel is the translation zero mode every MOTS stability operator carries — not a coincidence but the sphere’s freedom to be displaced. We reproduce this independently, by a second geometric route (surface-of-revolution principal curvatures) matched to the spectrum, so the GR leg of this argument rests on a published 2008 theorem rather than on our own symbolic algebra (`ams_stability.py`).

*Slicing caveat.* The clean *elliptic* ( $\partial_t$ -free) form is specific to the shear-free Newtonian slice with flat spatial sections. The AMS matter term for FRW is  $8\pi T_{ab}\ell^a k^b = 3H^2 + \dot{H}$ , so the *null-direction* (marginally-outer-trapped-tube, i.e. time-evolution) variation carries an  $\dot{H}$  that the within-slice shape operator does not: the two differ by exactly the horizon’s evolution term. It is the within-slice shape equation — the one governing whether the horizon *roughens* — that is elliptic and slaved; the  $\partial_t$  enters only the orthogonal question of how the horizon moves between slices.

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### 3. The premise: finite local dimension, and why $\alpha < d$ evades the area law

The two ingredients of §2 — a finite leg budget  $\kappa$  and a bond weight decaying slower than  $r^{-d}$  — are not exotic; each is weaker than something already standard.

Finite  $\kappa$  is unitarity. In a tensor-network representation of the horizon state, the weight on a leg attached to a site is the logarithm of the local bond dimension,  $\ln \chi$ , which a finite-dimensional local Hilbert space bounds. The per-site budget  $\sum_j w_{ij} = \kappa$  of (2.1) is therefore the statement that each horizon degree of freedom has a bounded amount of entanglement to distribute — a consequence of unitarity, not an assumption about geometry. The bond cut  $C(\Omega)$  is then the standard Ryu–Takayanagi / minimum-cut

upper bound on the entanglement entropy of  $\Omega$  (§6.1). Everything in §2.2–§2.4 uses only this budget and the *range* of the weights; it never uses the specific form  $1/r$ .

Why  $\alpha < d$  is allowed — the area-law theorems and their hypothesis. A natural objection is that entanglement entropy “always” obeys an area law, so a volume-law horizon is forbidden. It is not: the entanglement area-law *theorems* are theorems about *local* systems. In one dimension, gapped local Hamiltonians have area-law (there, constant) ground-state entanglement [11]; the higher-dimensional results [12] likewise assume short-range interactions and a spectral gap. Their load-bearing hypothesis is exactly locality — couplings that fall off fast enough that a region’s boundary dominates its entanglement — and that hypothesis fails precisely when the weights decay slower than  $r^{-d}$ : the same  $\alpha = d$  threshold at which our capacity crossover sits (§2.3) and at which long-range statistical mechanics stops being additive [10]. A volume-law entanglement entropy is thus not a violation of anything; it is the generic expectation once the connectivity is long-range. The premise (2.8) is the *negation* of the area-law theorems’ hypothesis, and the horizon — connected by gravity — is on the far side of it.

The reduction is therefore in the *strength* of what is assumed. The postulate replaced was “the horizon entropy counts its volume.” What remains is “the horizon degrees of freedom have finite local dimension and a longer-than-marginal-range coupling,” plus the state assumption of §6.1. The first is unitarity; the second is an inequality gravity satisfies with two decades of margin (§4); only the third is a genuine physical input, and it is the foundations paper’s, not ours.

#### 4. Gravity lands in the volume phase

Which side of the  $\alpha = d$  crossover is the cosmic horizon on? The bond weights of (2.1) encode which horizon degrees of freedom are entangled with which, and how strongly, as a function of separation. For a horizon whose microscopic correlations are set by gravity, that reach is Newtonian,  $w(r) \propto 1/r$ , so  $\alpha = 1$  — well below  $d = 3$ , deep in the volume basin (2.6), with a margin of two full units in the exponent. This is why the dichotomy of §2.4 is not delicate for the physical case: gravity does not sit near the crossover, and no  $O(1)$  uncertainty in the effective coupling could carry it across two decades to the area side.

That same long-rangedness is the object the foundations paper works with from the thermodynamic direction. There the horizon degrees of freedom carry a cooperative coupling  $J = \lambda_{\max}(W_{\text{grav}})$  — the largest eigenvalue of the very weight matrix  $W_{ij} \propto r^{-\alpha}$  of (2.1) — and  $\alpha < d$  is exactly the regime in which that coupling is *super-extensive* ( $\lambda_{\max} \propto N^{1-\alpha/d}$ ) and mean-field thermodynamics is exact [10]. So the network’s “ $\alpha < d \Rightarrow$  volume” of the present paper (the *count*) and the foundations paper’s “non-additive cooperativity  $\Rightarrow$  a bistable area $\leftrightarrow$ volume free energy  $F(m)$ , selected at the horizon’s birth” (the *selection*) are two readings of one fact: gravity is long-range at the horizon. The count fixes that the two branches are exactly {area, volume} and that  $\Delta \leq 1$ ; the selection fixes which branch a given horizon occupies — the grown cosmic horizon carried through the

ordering bifurcation onto the volume branch (a *tilted* pitchfork, whose absence of branch ambiguity is the generic feature of its universal unfolding [15]), the abruptly-formed black hole locked on the area branch. The two papers meet here: this one fixes the ceiling and the two available branches of the count, the foundations paper the dynamics that selects between them.

## 5. The edge correction and the infrared cutoff

### 5.1. The exact deficit law

Equation (2.6) is exact only for  $\rho \ll L$ . At finite  $\rho/L$  the outward fraction falls short, because the bond mass a site can place beyond  $\rho$  is limited by the ambient size. From (2.5), for  $\alpha < d$ ,

$$1 - s(\rho)/\kappa = D(x) = c x^{d-\alpha}, \quad x \equiv \rho/L. \quad (5.1)$$

At  $\alpha = 1$  no short-distance regulator is needed: Newton's theorem gives the exact angular average  $\langle |x - y|^{-1} \rangle = 1/\max(s, s')$ , and  $D(x)$  is a pure number. Solving the Sinkhorn problem in the radial reduction and refining the discretisation:

Table 3: Convergence of the deficit coefficient  $D(x)/x^2$  with the radial discretisation  $M$ , at  $\alpha = 1$  (exact Newton-kernel angular average).

| $M$  | $D(0.02)/x^2$ | $D(0.03)/x^2$ |
|------|---------------|---------------|
| 750  | 0.6738        | 0.6439        |
| 1500 | 0.6732        | 0.6732        |
| 3000 | 0.67308       | 0.67316       |
| 6000 | 0.67305       | 0.67315       |

$$c = \lim_{x \rightarrow 0} \frac{D(x)}{x^2} = 0.6730 \pm 0.0001,$$

with a fitted exponent 2.0018 (predicted  $d - \alpha = 2$ ). (Note that  $D/x^2$  drifts upward with  $x$  — 0.6731 at  $x = 0.02$ , 0.6817 at  $x = 0.20$  — so  $c$  must be read as the small- $x$  limit, not as the intercept of a power-law fit over a finite window.)

The exponent  $d - \alpha$  is confirmed independently at  $\alpha = 2$  (*deficit*  $\propto x$ ) and  $\alpha = 2.5$  ( $\propto x^{1/2}$ ). We record that  $c$  is *not*  $2/3 = 0.66667$ , from which it differs by 1%, well outside the discretisation error. We know of no closed form and do not assert one.

### 5.2. The effective deformation

With  $S(\rho) = \kappa \mathfrak{n} V(\rho)[1 - D(\rho/L)]$ ,

$$\Delta(x) = \frac{d \ln S}{d \ln \rho} - 2 = 1 - \frac{x D'(x)}{1 - D(x)} = 1 - \frac{(d - \alpha) c x^{d-\alpha}}{1 - c x^{d-\alpha}}. \quad (5.2)$$

At  $\alpha = 1$  (Figure 4):

Table 4: Effective deformation  $\Delta$  versus the ratio of horizon radius to network infrared cutoff,  $R_H/L$ , at  $\alpha = 1$  (eq. 5.2).

| $R_H/L$ | $\Delta$ |                                        |
|---------|----------|----------------------------------------|
| 0.001   | 1.0000   | network far exceeds the horizon        |
| 0.10    | 0.9864   | $L = 10 R_H$                           |
| 0.3125  | 0.8546   | $L = \text{comoving particle horizon}$ |
| 0.50    | 0.5522   | $L = 2 R_H$                            |

The deficit is one-signed:  $D \geq 0$  by (2.3), so the correction always pushes  $\Delta$  *downward*, toward the area law, and never above the ceiling. This is consistent with, and is the mechanism behind, (2.4).

### 5.3. $L$ is not fixed by the model

Expression (5.2) is a one-parameter family in  $L$ . Nothing in SEDE — not the CKN bound, not flatness, not the structure gate, not the birth-selection chain of the foundations paper’s §5 — determines the infrared cutoff of the horizon’s degree-of-freedom network. Adopting any finite  $L$  therefore introduces a continuously-sampled dark-sector parameter, which is precisely what the model’s central claim forbids (the cosmology paper’s §3: *the dark sector adds no fitted parameter*).

The only value of  $L$  consistent with the model’s own background is  $L \rightarrow \infty$ . The spatial slice of flat FRW is infinite; its flatness is already used, in the cosmology paper’s §2, to normalise  $\Omega_{DE}0$ . With  $L \rightarrow \infty$ ,  $D \rightarrow 0$  and

$$\Delta = 1 \quad \text{exactly.} \quad (5.3)$$

We are explicit about the status of this step, because it is where the value of  $\Delta$  actually enters. The argument above eliminates the would-be parameter; it does not derive its value. The step that fixes  $L$  is the identification of the cosmological spatial slice with the infrared cutoff of the entanglement network — an identification nothing in SEDE determines.  $\Delta = 1$  is therefore the *endpoint reading*: a cutoff choice made for consistency with the model’s own no-fitted-parameter claim, under which the pre-registered point carries no free parameter — and it remains the theory’s one postulate, restated, not a

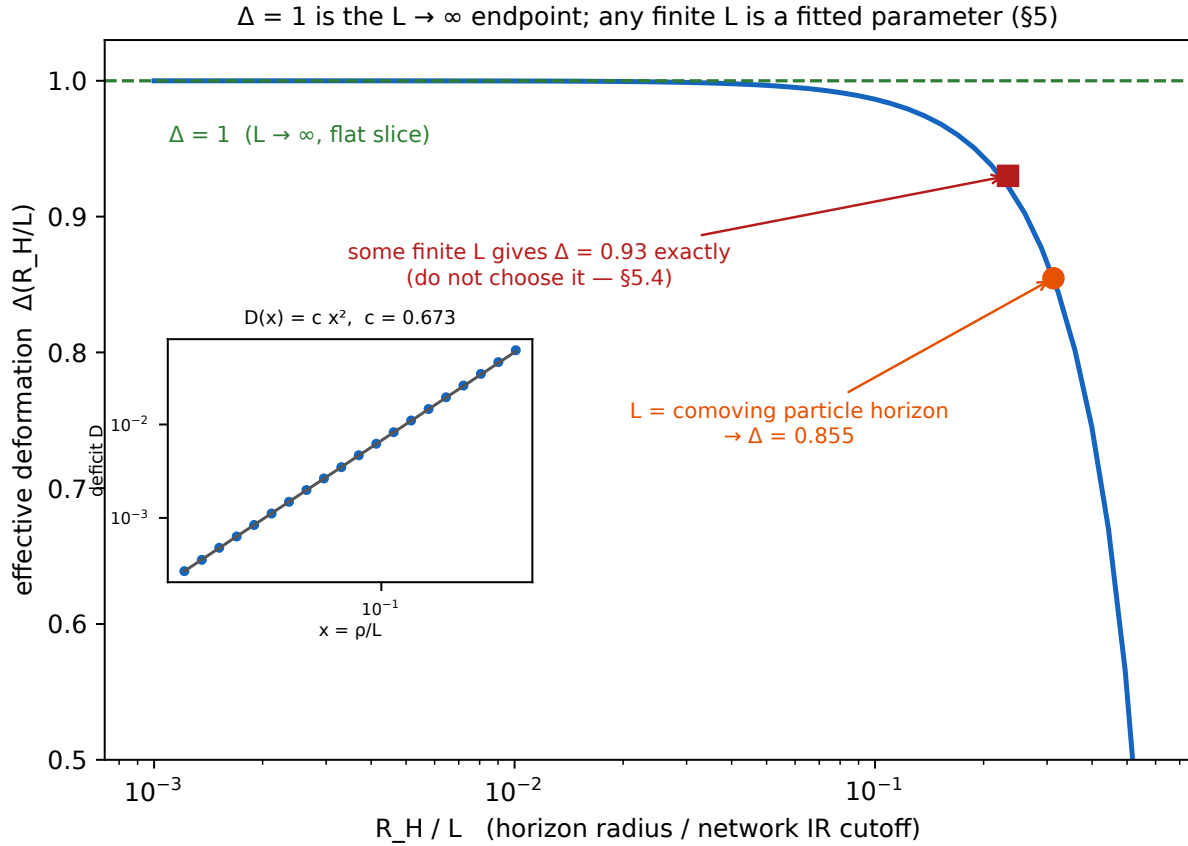


Figure 4.  $\Delta = 1$  and the infrared cutoff (§5). The effective deformation  $\Delta(R_H/L)$  rises to 1 as the network cutoff  $L \rightarrow \infty$ . Nothing in the model fixes  $L$ , so any finite value is a fitted dark-sector parameter — and choosing one is choosing the answer:  $L$  = the comoving particle horizon returns  $\Delta = 0.855$ , and some  $L$  reproduces the foundations paper’s profile value 0.93 exactly (§5.4 cautions against both). Identifying the flat, infinite spatial slice with the network cutoff takes  $L \rightarrow \infty$  and  $\Delta = 1$  — the endpoint choice of §5.3, which eliminates the parameter without deriving the value. Inset: the exact deficit law  $D(x) = c \cdot x^2$  at  $\alpha = 1$  (Newton kernel),  $c = 0.673$ .

theorem of this counting. (An earlier version of this paper presented  $\Delta = 1$  as “exact, with zero free parameters,” full stop; that phrasing conflated eliminating a parameter by a cutoff choice with deriving its value, and is corrected throughout.) The point value agrees with the foundations paper: an earlier  $\Delta \in [0.98, 1]$  window there, whose derivation rested on a logarithmic Edwards–Wilkinson roughening of the horizon surface, collapses to the point  $\Delta = 1$  once that roughening is removed — as it is both there and in §2.5 here. This paper now grounds that point value in the network capacity, as the  $L \rightarrow \infty$  endpoint. There was never a window that is both derived and free.

#### 5.4. A warning we place in the text deliberately

Setting  $L$  equal to the comoving particle horizon ( $\approx 3.2 R_H$ ) returns  $\Delta = 0.855$ . The profile-likelihood interval of the foundations paper’s §7 is  $\Delta = 0.93$  [0.83, 1.02]. There exists a choice of  $L$  returning 0.93 exactly.

We do not make it, and we recommend that no one does. A prediction is overdetermined only when every quantity entering it is fixed independently of the datum it is compared against. Here the only principle available fixes  $L = \infty$  and hence  $\Delta = 1$ ; any other value of  $L$  is selected by the answer. The same discipline applies to a second numerical coincidence in this programme: the volume well of the foundations paper’s free energy sits at  $m^* \approx 0.93$ , and  $m$  is an amplitude while  $\Delta$  is an exponent. Two coincidences at the same value, neither carrying information.

#### 5.5. Pre-registration

Table 5: Pre-registered predictions for the Barrow deformation  $\Delta$  and the conditions under which the framework is falsified.

| statement                        | value                                                                     | status                                                                                                                                |
|----------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| driven cosmic horizon            | $\Delta = 1$ exactly                                                      | pre-registered point; the $L \rightarrow \infty$ endpoint reading (a postulate, §5.3), no free parameter once the endpoint is adopted |
| equilibrium horizon (black hole) | $\Delta = 0$                                                              | the foundations paper’s §5(iii), unchanged                                                                                            |
| intermediate $\Delta$            | not produced by any gravitational ( $1/r, \alpha < d$ ) or local coupling | falsifies the framework (§2.4)                                                                                                        |
| $\Delta > 1$                     | impossible                                                                | falsifies the framework (§2.2)                                                                                                        |

DESI DR3 + Euclid reach a forecast  $\sigma(\Delta) \approx 0.09$  (Fisher, hence optimistic):  $\Delta = 1$  separates from  $\Delta = 0$  at a nominal  $\sim 11\sigma$  and from any intermediate value at  $\gtrsim 4\sigma$ .

## 6. From capacity to entropy: the state factor and the gate

Everything in §2 concerns the *capacity*  $C(\Omega)$  — the bond-cut upper bound on the entanglement entropy of a region. Turning a capacity into the horizon entropy  $S$  that



enters the cosmology (the cosmology paper's  $\rho_{\text{DE}} = T_{\text{AH}} s_{\text{grav}}$ ) takes one further ingredient, and it is worth isolating exactly what it is and what it is not.

### 6.1. Attainment is the state question

The bond cut  $C(\Omega) = \sum_{i \in \Omega, j \notin \Omega} w_{ij}$  is the Ryu–Takayanagi/minimum-cut *bound*:  $S(\Omega) \leq C(\Omega)$ , with equality only for a state that saturates it. So the whole of §2 — the theorem  $\Delta \leq 1$ , the {area, volume} dichotomy, the  $\alpha = d$  crossover — is *state-independent*; but the identification  $S = C$ , which fixes the *value* of the entropy rather than its scaling, is not. It holds when the horizon degrees of freedom are maximally entangled (thermalised, Page-saturated), so that every bond across the cut carries its full  $\ln \chi$ . That is precisely the *state* half of the postulate, argued independently in the foundations paper (route A: Sachdev–Ye–Kitaev (SYK) / maximal scrambling) and untouched here. We are explicit that this is the one thing §2 does not remove: the network fixes the *count* — which degrees of freedom a region's entropy may draw on — while the maximally-scrambled state fixes that they are all in fact used. §2 makes the count a theorem; the value of the entropy still rests on the state.

The saturating state is not exotic: it is the tracial state of the site algebra. The assumption can be sharpened from “some state saturates every cut” to an identification internal to the model. Each site carries a finite Hilbert dimension  $e^\kappa$ , so the site algebra of a region  $\Omega$  is a finite-dimensional factor with  $\dim \mathcal{H}_\Omega = e^{\kappa|\Omega|}$ . Its maximally mixed (tracial) state assigns

$$S(\Omega) = \log \dim \mathcal{H}_\Omega = \kappa|\Omega| \quad \text{for every } \Omega,$$

i.e. it attains the capacity bound of §2.2 identically, region by region — not as a fine-tuned configuration but as the maximum-entropy state of the very algebra whose capacity §2 bounds. The “cut-saturating state” and the “maximally scrambled state” are therefore the same object described from two sides, and  $S = C$  holds in it by construction.

Two qualifications keep this honest. First, saturation of *every* cut requires the global state to be mixed: a pure state obeys  $S(\Omega) = S(\Omega^c)$  and is capped by  $\min(\kappa|\Omega|, \kappa|\Omega^c|)$ , so it cannot saturate both a region and its complement. For a horizon-bounded region mixedness is the physical situation — the exterior purifies the interior — and it is also the situation in the crossed-product description, whose maximal-entropy state is likewise maximally mixed. Second, this identifies the saturating state; it does not prove that a given *physical* horizon is in it. What it removes is the suspicion that the saturating state is a special construction invented to reach  $\Delta = 1$ : it is the flat state of the model's own algebra, and the remaining question — whether horizon dynamics drives the state to it — is the scrambling question already stated, not an additional one.

### 6.2. The entropy density and the occupancy gate

On the volume branch ( $\alpha < d$ ) with the saturating state,  $S(\rho) = C(\rho) = \kappa \mathfrak{n} V(\rho)$ , so the entropy density is the constant

$$s_0 = \kappa n = \kappa / \ell_P^3 \quad (6.1)$$

of eq. (2.7) — the cosmology paper’s  $s_{\text{grav}}$ , legs per Planck cell. A horizon does not, however, carry its full volume capacity at every epoch: the capacity is *populated* as structure drives it. Writing  $f_{\text{sat}} \in [0, 1]$  for the activated fraction,

$$S = f_{\text{sat}} s_0 V, \quad (6.2)$$

the driven limit  $f_{\text{sat}} \rightarrow 1$  is the volume law ( $\Delta = 1$ ) and the undriven limit  $f_{\text{sat}} \rightarrow 0$  is the area baseline ( $\Delta = 0$ ) — the state-dependent count of the foundations paper’s §7.1, now with  $s_0$  supplied by the network rather than posited. The gate is not a new function: minimal absorption kinetics — each deposit activates a fraction of the *remaining* inactive capacity,  $df = \gamma(1 - f) d(D^2)$  — integrate to  $f_{\text{sat}} = 1 - e^{-\gamma D^2}$ , exactly the cosmology paper’s gate (the foundations paper’s §7.1a).

### 6.3. Occupancy is not the order parameter (a correction to Fig. 3 of the foundations paper)

It is tempting to identify  $f_{\text{sat}}$  with the order parameter  $m$  of the area $\leftrightarrow$ volume free energy  $F(m)$  of the foundations paper’s §5. They are different variables, and their dynamics part company at the endpoint. The order parameter is a *free-energy* coordinate: it relaxes by gradient flow,  $\partial_\tau m = -\Gamma F'(m)$ , a *branch-selecting* dynamics fixed by the shape of  $F$  and driven by the pitchfork at the horizon’s birth. The occupancy is a *cumulative* coordinate: it obeys the gate kinetics  $\partial_x f = \gamma(1 - f)$  (§6.2), whose rate vanishes as  $f \rightarrow 1$  — a monotone filling that can never overshoot and carries no barrier. A variable whose own equation of motion switches off at the endpoint it is approaching cannot be the order parameter of a double-well potential. The division of labour is therefore clean: the *branch* — which well the horizon sits in — is fixed at birth by  $m$  (the foundations paper’s §5 iii);  $f_{\text{sat}}$  is the subsequent *occupancy* of the already-selected volume capacity, not a second branch selector. Insofar as the presentation of Fig. 3 of the foundations paper invites its landscape to be read against  $f_{\text{sat}}$ , the landscape variable is the order parameter  $m$ ;  $f_{\text{sat}}$  is the distinct occupancy coordinate of (6.2).

### 6.4. The loop to the cosmology paper

Equations (6.1)–(6.2) close the chain from the count to the observable with no fitted dark-sector parameter. The entropy density  $s_0 = \kappa n$  is fixed by unitarity (legs per Planck cell); the exponent  $\Delta = 1$  by §2 (the volume branch) and §5 (the flat slice,  $L \rightarrow \infty$  — the endpoint choice); the gate  $f_{\text{sat}}$  by the derived kinetics of §6.2. These are exactly the two dark-sector inputs the cosmology paper’s horizon-fluid ansatz  $\rho_{\text{DE}} = T_{\text{AH}} s_{\text{grav}}$  requires — the volume-law entropy density and its occupancy — so the cosmology paper’s background is reproduced with every dark-sector quantity now grounded in the network count. What is *not* discharged, and we say so plainly, is the state assumption of §6.1: §2 makes the ceiling and the dichotomy theorems, but the value of the horizon entropy is

attained only for the maximally-scrambled state carried over from the foundations paper. That single assumption is the honest residue of the capacity→entropy step.

## 7. The residue

Table 6: The assumption ledger: status of each component of the volume-law input before this paper (the foundations paper’s §8) and after the reduction.

| item                                                 | status before (the foundations paper’s §8)                                                                | status now                                                                                                                             |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| state (maximal entanglement)                         | first-principles (route A / SYK)                                                                          | unchanged                                                                                                                              |
| form ( $S \propto V$ )                               | reduces to thermalisation                                                                                 | unchanged                                                                                                                              |
| scale ( $\rho_{\text{DE}} \sim \rho_{\text{crit}}$ ) | CKN bound + flatness                                                                                      | unchanged                                                                                                                              |
| count (bulk vs boundary)                             | open — the residue                                                                                        | reduced (§2, §5): binary {0, 1} given the network model; $\Delta = 1$ is the $L \rightarrow \infty$ endpoint choice, still a postulate |
| geometric ceiling $\Delta \leq 1$                    | postulate in effective form (cosmology paper, §8.1); reduced to bound-only in the foundations paper, §7.1 | theorem (2.3)–(2.4)                                                                                                                    |
| $\Delta = d_{\text{H}} - 2$ (fractal horizon)        | assumed (cosmology paper, §8.1); withdrawn in the foundations paper                                       | withdrawn (§2.5)                                                                                                                       |
| roughening class (EW vs KPZ)                         | conditional prediction (foundations paper, pre-fold)                                                      | no such classification exists                                                                                                          |
| transverse diffusivity coefficient                   | scoped open coefficient (foundations paper, §5 iv)                                                        | not defined (§2.5); the $\epsilon$ -lag layer it fed is retracted in the foundations paper (§6)                                        |
| —                                                    | —                                                                                                         | <i>new</i> : finite leg budget $\kappa$ (unitarity)                                                                                    |
| —                                                    | —                                                                                                         | <i>new</i> : $w(r)$ decays slower than $r^{-d}$                                                                                        |
| —                                                    | —                                                                                                         | <i>new</i> : the horizon dof are a network                                                                                             |

| item | status before (the foundations paper's §8) | status now                                                                    |
|------|--------------------------------------------|-------------------------------------------------------------------------------|
| —    | —                                          | <i>new</i> : the infinite-cutoff identification $L \rightarrow \infty$ (§5.3) |

Four premises remain, and they are weaker than the one they replace. The leg budget is unitarity. The bond-weight condition is an inequality, not a functional form, and gravity satisfies it with two decades of margin in the exponent. The infrared-cutoff identification is a stated endpoint choice — the residue of the original postulate, relocated and made explicit rather than eliminated (§5.3). The state premise is untouched and independently argued. What has gone is the bulk-versus-boundary count *as a free binary* — the item both previous papers name as irreducible is now a binary forced to  $\{0, 1\}$  by the network's range dichotomy — together with the fractal reading of  $\Delta$  and the roughening layer built on it.

We claim a reduction, not a proof. The bond cut is a capacity, an upper bound on entanglement entropy; its attainment is the state question, and the identification of the horizon degrees of freedom with a network of the assumed connectivity is a model of the horizon, not a derivation of one. What the reduction buys is that the remaining conditionality now sits in one place, is stated as an inequality, and is decided — for the one horizon we have — by a measurement already scheduled.

## 8. Discussion

Status: this is a complexity/connectivity layer, not a proven entropy derivation (governing note). Read the construction below as a model of coarse-grained horizon *connectivity*, decoupled from any claim to have derived a gravitational entropy or a stress tensor. Three points govern. (i) The volume-scaling quantity is a structural complexity  $C_{\text{str}}$ , not the generalized entropy  $S_{\text{gen}}$ : the bond-cut capacity bounds how many degrees of freedom a region's entropy *draws on*, which need not equal  $A/4G + S_{\text{out}}$ . (ii) A horizon-tracking CKN cutoff gives an *area-like*  $\Delta = 0$  (density  $\propto H^2$ ), not the volume-like  $\Delta = 1$ ; the volume count is the *non-default* identification, allowed by long-rangedness ( $\alpha < d$ ) but not forced, and the one concrete de Sitter model (double-scaled SYK) currently reads as area — so  $\Delta = 1$  remains the empirical bet of the cosmology paper, tested by DESI DR3 + Euclid, not a theorem of this counting. (iii) The bridge to cosmology is an *empirical ansatz*, not a derivation:

$$C_{\text{str}} \longrightarrow q_R = \sigma_R^2 \longrightarrow u(q_R),$$

i.e. the complexity is identified with the model-owned filtered matter variance  $q_R = \sigma_R^2$  ( $R = 8 \, h^{-1} \text{Mpc}$ ) that gates the FPAB-SEDE background — and its ensemble and renormalization are *not* specified here. We do not infer the cosmological amplitude  $\mu$ ,

the filter scale  $R$ , the normalization  $q_*$  or the braiding fraction  $b$  from node counting; those are fixed by the finite cubic-KGB action and the data (the cosmology paper), and all cosmological comparisons below should be read against that action's predictions, not the earlier smooth-fluid solution.

The de Sitter holography question, addressed constructively. The residue lives in de Sitter holography [16]. The foundations paper reduced the count to its sharpest form: in the crossed-product (Type II<sub>1</sub>) description of the de Sitter static patch, the *value* of the maximal horizon entropy is a trace calibration — a free normalisation the algebra does not fix — so “the eternal-patch ceiling is the area” is an Einstein-saddle calibration, not an algebraic output, and the FRW-patch calibration is left open as “the count asked algebraically.” This paper answers it constructively, without solving the operator-algebraic problem: the trace is fixed by the network —  $\kappa$  legs per Planck cell — and the count that trace delivers is *volume*, because the horizon's connectivity is long-range ( $\alpha < d$ ). We do not claim to have computed  $\dim \mathcal{H}(\text{static patch})$  non-perturbatively; we claim that the quantity the II<sub>1</sub> calibration stands in for — how many degrees of freedom a horizon region's entropy draws on — is a bond count, and that bond count is the enclosed volume for any gravitational horizon. Consistently with the foundations paper, the algebra *type* is not the discriminator (an accelerating attractor is II<sub>1</sub>-compatible either way); the calibration is, and the calibration is the leg budget.

What is not settled. One thing, stated plainly — the list is shorter than in earlier drafts, and we record what moved. The  $\alpha = d$  crossover of §2.3 previously rested on a dominance argument that was correct term by term but interchanged the  $\rho \rightarrow \infty$  and continuum limits without justification; Proposition 2.1 now replaces that interchange with a two-sided bound at finite  $(\ell, \rho, L)$ , uniform in the site position, whose error term is explicitly  $O((\rho/L)^{d-\alpha})$  and is verified numerically. The derivation gap is closed; what remains there is only that the proved constant is not sharp (loose by  $\approx 17$  at  $\alpha = 1$ ), which affects no conclusion. What is *not* settled is the identification  $S = C$ , the state question (§6.1): the network gives the count, the maximally-scrambled state gives that the count is attained. That assumption is route A of the foundations paper, and it is the honest residue of the whole programme — now a single question, and one shared with area-law black holes (a black hole carries the same maximally-entangled horizon state; what differs is that it is undriven, so its capacity is the boundary count, §6.2). The open content of SEDE is thereby relocated, in its entirety, from the *count* to the *state*. That state question is itself now narrower than it was: §6.1 identifies the saturating state as the tracial state of the site algebra — the flat state of the model's own algebra, which attains the capacity bound for every region identically — so what remains is not “does a saturating state exist” but “is a physical horizon driven to it,” which is the scrambling / Page-curve question the foundations paper takes up. We do not claim that last step is closed here.

Falsifiability. Because the count is discrete for a gravitational horizon (§2.4), the prediction is a *point*,  $\Delta = 1$ , not a fitted exponent: an intermediate  $\Delta$  is not a universality class to be selected but a refutation, and DESI DR3 + Euclid forecast  $\sigma(\Delta) \approx 0.09$  (Fisher) — separating volume from area at a nominal  $\sim 11\sigma$  and from any intermediate value

at  $\gtrsim 4\sigma$ . The companion black-hole channel ( $\Delta = 0$  for equilibrium horizons, §6.2) makes the framework falsifiable on a *second*, independent horizon: an isolated volume-law black hole would mean the deformation is universal, refuting the state-dependent, driven-versus-equilibrium count. Resting the count on a network makes SEDE more falsifiable, not less.

## 9. Conclusion

We set out to reduce the one input the foundations paper could not — whether the cosmic horizon’s entropy counts its enclosed bulk or its boundary. Modelling the horizon degrees of freedom as a network with a finite leg budget and long-range bonds, we found the *ceiling* is a theorem and the count binary: the bond-cut capacity bounds  $\Delta \leq 1$  identically; the capacity is the enclosed volume for every coupling of range  $\alpha < d$  and the boundary area for  $\alpha > d + 1$ , with gravity’s  $1/r$  reach ( $\alpha = 1$ ) deep on the volume side. The value  $\Delta = 1$  is not a theorem of this counting: it is the  $L \rightarrow \infty$  endpoint of the one free scale — the infrared cutoff, which nothing internal fixes — reached by identifying the flat, infinite spatial slice with the network cutoff, a stated choice under which the pre-registered point carries no free parameter. The fractal reading of  $\Delta$  is unnecessary (the ceiling is a finite-dimension bound) and, we showed, unavailable (no minimum-cut surface exists for a long-range network, hence no Hausdorff dimension and no roughening) — which retires, across the companion papers, the roughening layer once built on it. What remains is strictly weaker than the postulate it replaces: finite local dimension, a range inequality gravity satisfies with room to spare, the infinite-cutoff identification, and a maximally-scrambled state. The last two are the residue — a *state* question and an *endpoint* choice, no longer a free counting — the same questions area-law black holes now answer, and the same measurement,  $\Delta$  to a forecast  $\sigma \approx 0.09$  at DESI DR3 + Euclid, that decides them for the one horizon we have.

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## Reproducibility

Every number above is produced by the accompanying code from a single entry point, with validation assertions at each stage:

- `src/mincut_membrane.py` — maximum-flow minimum cut on the lattice ball; membrane-thickness and shell-fraction diagnostics (§2.5, tables i–ii).
- `src/capacity_radial.py` — spherical reduction with the exact angular average; the  $\alpha$ -crossover table of §2.3 and the large- $\rho$   $C(\rho)/\rho^2$  diagnostic (marginal-log at  $\alpha = d+1$  vs clean area at  $\alpha > d + 1$ ), asserted.
- `src/outward_fraction_bound.py` — the finite-size outward-fraction bound of Proposition 2.1: fits the capacity-deficit exponent against the predicted  $d - \alpha$  and checks every measured deficit against the proved bound  $K(\rho/L)^{d-\alpha}$ , asserted.



- `src/edge_deficit.py` — Newton-kernel solve at  $\alpha = 1$ ; the deficit law and the convergence of  $c$  (§5.1).
- `src/horizon_constraint.py` — symbolic expansion of the apparent-horizon condition, eq. (2.9); verified against the exact mean curvature of a translated sphere.
- `src/ams_stability.py` — external-standard check: eq. (2.9) is the Andersson–Mars–Simon MOTS stability operator on the FRW round sphere. Independent (surface-of-revolution) derivation of  $-(1/\bar{R})(\nabla_S^2 + 2)$ , the published spectrum  $\ell(\ell + 1) - 2$  with the  $\ell = 1$  translation zero mode, and the AMS matter term  $8\pi T_{ab}\ell^a k^b = 3H^2 + \dot{H}$  that isolates the slicing caveat.
- `run_all.py` — single entry point; runs every stage with validation assertions.
- `make_figures.py` — regenerates the four figures from the same `src/` computations (Figs 1–4  $\rightarrow$  `output/count_fig{0-3}_*.png`).

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## Data availability statement

The code that supports the findings of this article is openly available at <https://github.com/spsingularity/sede-count>, with a tagged release archived at Zenodo, DOI [10.5281/zenodo.21525523](https://doi.org/10.5281/zenodo.21525523). Every quantitative claim is reproduced from a single entry point (`run_all.py`, also wired into the repository-wide `reproduce_all.py` as the `netcount` stage), each stage carrying validation assertions that all pass, and the four figures regenerate from the same computations via `make_figures.py`. No observational data were generated or analysed beyond the standard public inputs cited in the companion papers.

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## Competing interests

The author, an independent researcher, declares no competing interests.

## Ethics statement

Not applicable. This work is purely theoretical and involved no human participants, human data or tissue, and no animal subjects.

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As sole author, S. Pandev conceived and carried out the study, performed and independently verified all analyses, and wrote the manuscript, taking full responsibility



for its content. Artificial-intelligence tools (large language models) assisted with drafting and editing the manuscript, developing and cross-checking the analysis code, and literature searches; no AI tool is an author.

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